

# Image in Software User Interface Design

Guidelines for meaningful, usable, and accessible interface images

Software User Interface Design

## Section 1

### Context and Scope

# Topic: Image in SUID

**Core idea:** In user interface design, an image is not decoration first; it is a **communication element**.

Images can:

- support understanding and recall,
- make abstract information concrete,
- guide navigation,
- communicate object appearance,
- improve accessibility when used correctly,
- create confusion, delay, and distraction when used carelessly.

**Reference base:** Wilbert O. Galitz, *The Essential Guide to User Interface Design*, 3rd ed., Step 11: “Create Meaningful Graphics, Icons, and Images,” especially the section **Images**, pp. 671–675.

# Flow of the Presentation

- ① Why images matter in interface design
- ② Where images fit among graphics and multimedia
- ③ Purposes and types of interface images
- ④ Design principles for images
- ⑤ Image size, loading, and performance
- ⑥ Image color and file format decisions
- ⑦ Accessibility and internationalization
- ⑧ Related image techniques: image maps and photographs
- ⑨ Practical checklist and conclusion

## Section 2

### Why Images Matter

# Images as Interface Communication

Images are part of the visible language of an interface.

A good image can:

- communicate faster than text in some situations,
- make an interface easier to learn,
- reduce dependence on language,
- help users recognize objects and actions,
- support navigation through visual cues.

However, a poor image can:

- mislead the user,
- increase page clutter,
- slow download and response time,
- reduce accessibility,
- distract from the main task.

# Key Principle

*Use images only when they serve a clear user-interface purpose.*

Images should **supplement** the interface message, not dominate it.

They should be selected or designed according to:

- user goals,
- task context,
- recognizability,
- legibility,
- accessibility,
- performance constraints.

## Section 3

### Images within Graphics and Multimedia



# From Graphics to Images

In SUID, images are part of a broader set of graphical and multimedia elements.

| Element     | Main purpose in UI                           | Example                              |
|-------------|----------------------------------------------|--------------------------------------|
| Icons       | Represent objects, actions, status, warnings | Save, delete, warning                |
| Graphics    | Supplement text and support navigation       | Site overview, category illustration |
| Images      | Convey visual meaning and recognition        | Product image, navigation image      |
| Photographs | Show realistic appearance                    | Person, product, location            |
| Diagrams    | Show structure, relationship, or process     | Flowchart, organization chart        |
| Video       | Show motion or change over time              | Demonstration, tutorial              |

# Purposes of Graphics Containing Images

Galitz classifies graphics on Web/interface pages by purpose:

| Purpose          | Meaning                           | UI example                         |
|------------------|-----------------------------------|------------------------------------|
| Navigational     | Identifies links or routes        | Image button leading to a section  |
| Representational | Shows something mentioned in text | Product picture beside description |
| Organizational   | Shows relationships among items   | Site map or category map           |
| Explanative      | Explains how something works      | Diagram of a process               |
| Decorative       | Adds visual appeal or emphasis    | Background pattern or separator    |

**Important:** Decorative images must not interfere with task performance.

## Section 4

### Design Principles for Images

# Guideline 1: Ensure Images Convey Intended Messages

Users and designers may interpret the same image differently.

- Users often prefer **familiar** images.
- Designers may prefer **artistic** or visually impressive images.
- The best image is the one that communicates correctly to the target users.

**Design implication:** Test images with real users whenever possible.

**Questions to ask:**

- What should the image communicate?
- Do users understand it without explanation?
- Does it support the task?
- Could text communicate the same thing more clearly?

## Guideline 2: Use Standard Images

Use standard or already-tested images whenever possible.

Benefits:

- improves consistency,
- reduces learning effort,
- supports user expectations,
- avoids unnecessary invention,
- reduces interpretation errors.

Sources of standard images may include:

- platform UI guidelines,
- organizational design systems,
- industry standards,
- ISO and other standards bodies,
- existing application conventions.

## Guideline 3: Emulate Real-World Objects

Images that resemble real-world objects are easier to understand.

Examples:

| Interface image     | Real-world metaphor | Why it works                         |
|---------------------|---------------------|--------------------------------------|
| Button-like image   | Physical button     | Users understand it can be pressed   |
| Folder image        | Office folder       | Suggests storage or grouping         |
| Shopping cart image | Store cart          | Suggests items selected for purchase |
| Trash bin image     | Waste container     | Suggests deletion or removal         |

**Caution:** A real-world metaphor must be familiar to the target user group.

## Guideline 4: Use Images Consistently

An image should have **one meaning** throughout the system.

Avoid:

- different images for the same meaning,
- same image for different meanings,
- inconsistent style across pages,
- changing interactive behavior of similar-looking images.

Consistency helps users build a reliable mental model of the interface.

## Guideline 5: Produce Legible Images

An image must be easy to identify.

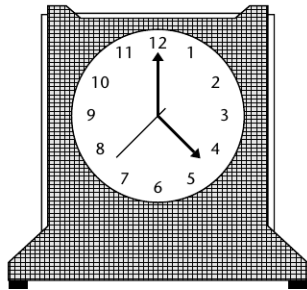
Legibility depends on:

- contrast with background,
- image complexity,
- image size,
- viewing distance,
- screen resolution,
- color choices,
- amount of detail.

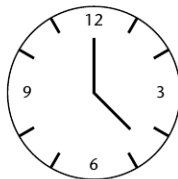
**Rule of thumb:** Use the minimum amount of detail needed for recognition.



# Visual Placeholder: Legibility and Detail



POOR



GOOD

**Figure 11.5:** Avoid excessive detail in icon design.

Figure 1: Figure 11.5: Avoid excessive detail in icon design. Source: Galitz, Step 11, printed p. 662.

## Guideline 6: Provide Descriptive Text or Labels

Images are not always self-explanatory.

Provide:

- visible labels for important images,
- alternate text for screen readers,
- captions where useful,
- longer descriptions for complex images.

This improves:

- comprehension,
- learning,
- recall,
- accessibility,
- usability when images fail to load.

# Descriptive Text: Why It Matters

Alternate/descriptive text helps different users:

| User situation       | Benefit of descriptive text                   |
|----------------------|-----------------------------------------------|
| Vision-impaired user | Screen reader can communicate image content   |
| Slow connection      | User understands image before it fully loads  |
| Images disabled      | Interface remains meaningful                  |
| Search/indexing      | Text content is easier to process than images |
| New user             | Label reduces guessing and errors             |

## Section 5

# Navigational and Decorative Images

# Distinguish Navigational Images from Decorative Images

Users must know which images are interactive.

A navigational image should clearly look clickable.

Possible cues:

- raised or button-like appearance,
- underlined label near the image,
- clear hover/focus state,
- consistent placement in navigation area,
- meaningful alternate text.

Avoid forcing users to move the pointer over every image just to discover what is clickable.

# Navigational vs Decorative Images

| Aspect        | Navigational image                       | Decorative image                            |
|---------------|------------------------------------------|---------------------------------------------|
| Purpose       | Takes user to content or performs action | Adds visual appeal/emphasis                 |
| Interaction   | Clickable or selectable                  | Usually not clickable                       |
| Required cue  | Must visibly indicate interactivity      | Should not look like a control              |
| Accessibility | Needs meaningful alt text                | May use empty alt text if purely decorative |
| Risk          | Hidden links, missed actions             | Distraction, clutter                        |

# Common Problem: Image Looking Like an Advertisement

Important functional images should not look like:

- banner ads,
- irrelevant decorations,
- promotional graphics,
- visual noise.

If users think a useful image is an advertisement, they may ignore it.

**Design goal:** Important images should look integrated with the interface and task.

## Section 6

### Minimizing Images



# Guideline 7: Minimize the Number of Images

More images usually mean:

- slower loading,
- more visual clutter,
- more cognitive effort,
- more competition for attention.

Use images only when they:

- add value,
- improve clarity,
- support recognition,
- help navigation,
- explain what text cannot explain efficiently.

## Guideline 8: Minimize Image Size

Large images increase waiting time and reduce responsiveness.

Galitz gives classic Web-performance targets:

| Image quantity    | Recommended limit in Galitz |
|-------------------|-----------------------------|
| Single image      | about 5 KB                  |
| Total page images | about 20 KB                 |

These numbers reflect older Web constraints, but the principle remains valid:  
*Keep images as small as possible while preserving necessary quality.*

# Practical Modern Interpretation

Even with faster networks, image size still matters because of:

- mobile data limits,
- low-bandwidth users,
- server cost,
- page load performance,
- energy usage,
- user patience.

Modern practice:

- compress images,
- use responsive image sizes,
- lazy-load below-fold images,
- use thumbnails for previews,
- avoid oversized source files.

## Guideline 9: Use Thumbnails

A **thumbnail** is a small, low-size preview of a larger image.

Use thumbnails when:

- many images must be shown,
- users may not need every full-size image,
- full images are large,
- quick scanning is important.

Best practice:

- show thumbnail first,
- link to high-quality full image,
- inform user about the full image size if relevant.

## Section 7

# Image Animation and Extraneous Images

## Guideline 10: Minimize Image Animation

Animated images should be used only when they serve a real purpose.

Useful purposes may include:

- showing change over time,
- demonstrating a process,
- giving meaningful feedback,
- attracting attention to a critical event.

Avoid animation for mere decoration because it can:

- distract users,
- slow loading,
- annoy users,
- create accessibility problems.

# Avoid Extraneous or Gratuitous Images

A gratuitous image is an image that does not support the interface goal.

Problems caused:

- visual clutter,
- slower interaction,
- confusion about what matters,
- reduced professionalism,
- poor accessibility.

**Test:** If removing the image does not reduce understanding or task success, the image may not be necessary.

## Section 8

### Color and File Format



# Guideline 11: Minimize Number of Colors

Reducing colors can reduce file size.

However:

- too few colors may degrade the image,
- rich photographic images need more color detail,
- flat graphics may work well with fewer colors.

**Design goal:** retain enough quality for recognition while reducing unnecessary file weight.

## Guideline 12: Choose the Appropriate Format

Galitz emphasizes two common Web formats of the time:

| Format | Best for                                  | Strength                               | Limitation                                                         |
|--------|-------------------------------------------|----------------------------------------|--------------------------------------------------------------------|
| GIF    | Flat colors, simple graphics, backgrounds | Small files, supports simple animation | Limited to 256 colors                                              |
| JPEG   | Photographs and rich tonal variation      | Good for complex color images          | Can introduce artifacts; usually larger than GIF for flat graphics |

# Modern Format Extension

For current UI work, also consider:

| Format | Best for                               | Notes                                                         |
|--------|----------------------------------------|---------------------------------------------------------------|
| PNG    | UI graphics, transparency, screenshots | Lossless, often larger than JPEG                              |
| SVG    | Icons, logos, diagrams                 | Scalable vector format; excellent for resolution independence |
| WebP   | Web images, photos, graphics           | Good compression; broad browser support                       |
| AVIF   | High-compression modern images         | Very small files but compatibility must be checked            |

Use the format that best matches the image purpose and platform support.

# GIF vs JPEG: Classic Decision Table

| Image type             | Prefer GIF      | Prefer JPEG |
|------------------------|-----------------|-------------|
| Simple line drawing    | Yes             | No          |
| Flat-color graphic     | Yes             | No          |
| Logo with few colors   | Yes / PNG / SVG | No          |
| Photograph             | No              | Yes         |
| Smooth color gradients | Usually no      | Yes         |
| Simple animation       | Yes             | No          |
| Detailed natural scene | No              | Yes         |

## Section 9

# Layout and Placement

## Guideline 13: Limit Large Images Above the Page Fold

Do not fill the first visible screen only with a large image.

Why?

- Users may not realize more content exists below.
- Important text may be hidden.
- The page may appear empty or purely promotional.
- Users may not scroll.

Better approach:

- combine image with meaningful heading/text,
- keep primary task visible,
- provide clear scroll cues,
- place large supporting images lower on the page or behind a thumbnail.

# Guideline 14: Use Background Images Sparingly

Background images can harm usability.

Problems:

- reduce text readability,
- slow loading,
- create visual noise,
- distract from content,
- cause contrast/accessibility failures.

If background images are used:

- keep them simple,
- use low contrast behind text,
- ensure strong text-background contrast,
- avoid busy patterns,
- test on different displays.

# Guideline 15: Reuse Images

Repeated images can be cached by browsers or applications.

Benefits:

- faster loading after first use,
- visual consistency,
- reduced design effort,
- easier maintenance.

Examples:

- repeated navigation icons,
- consistent section markers,
- common product placeholders,
- standard status illustrations.



## Section 10

# Accessibility

# Images and Accessibility

Screen-review utilities cannot directly reveal images to visually impaired users.

Therefore, important images need text equivalents.

| Image type               | Accessibility treatment                      |
|--------------------------|----------------------------------------------|
| Informational image      | Provide meaningful alt text                  |
| Complex diagram          | Provide short alt text + long description    |
| Navigational image       | Describe destination/action                  |
| Decorative image         | Empty alt text may be appropriate            |
| Image with embedded text | Avoid when possible; provide equivalent text |

# Long Descriptions

Some images cannot be explained in one short phrase.

Examples:

- charts,
- diagrams,
- maps,
- infographics,
- screenshots with many elements.

For these, provide:

- short alt text identifying the image,
- nearby explanation or caption,
- separate long-description page if necessary.

## ***Web Page Accessibility Design***

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- Pages:
  - Provide a simple and consistent layout.
  - Place important information at the page top.
  - Provide simple backgrounds contrasting well with the text.
  - Provide a “Skip to Main Content” link at the top of each page.
  - Structure articles with two or three levels of headings.
  - End sentences, headings, and list items with punctuation.
  - Provide frame titles.
  - Provide user adjustable font sizes and styles, colors, graphical attributes, and volume.
  - Avoid blinking or constantly changing elements.
- Controls:
  - Provide large buttons.
- Links:
  - Provide fully descriptive headings.
  - Separate consecutive links with a dividing character.
- Tables, frames, and columns:
  - Use sparingly.
  - Provide alternate ways to access items contained within tables.
- Images:

## Section 11

# Internationalization

# Image Internationalization

Images may reduce the need for translation, but they are not automatically universal.

Consider:

- cultural meanings,
- religious symbols,
- gestures,
- color associations,
- number meanings,
- social norms,
- reading direction,
- local objects and metaphors.

An image familiar in one culture may be meaningless or offensive in another.

# Internationalization Checklist

| Question                                         | Why it matters                                   |
|--------------------------------------------------|--------------------------------------------------|
| Is the object familiar to the target culture?    | Recognition depends on user experience           |
| Does the image contain text?                     | Text requires translation/localization           |
| Does the color have cultural meaning?            | Colors may signal different emotions or meanings |
| Does the gesture/symbol have local implications? | Gestures can be offensive or misunderstood       |
| Is the metaphor universal?                       | Some metaphors are culture-specific              |

## Section 12

### Image Maps



# What Is an Image Map?

An **image map** is a single image divided into clickable regions, where each region links to different content.

Use image maps mainly for navigation when the graphical structure itself is meaningful.

Examples:

- geographic map with clickable regions,
- building floor plan,
- product parts diagram,
- visual site overview.

# Image Maps: Advantages and Disadvantages

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## Advantages

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- Can match the user's mental model
- Can show meaningful structure
- May aid conceptualization
- Can be faster than many separate images

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## Disadvantages

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- Consumes significant screen space
- Hot spots may not be obvious
- Selected location may not be clear
- Search and accessibility issues may occur

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# Image Map Guidelines

Use image maps with caution.

If used:

- make clickable boundaries obvious,
- provide strong visual cues,
- show selected/visited states where possible,
- provide equivalent text links,
- ensure keyboard accessibility,
- provide alt text for regions,
- avoid making users guess where to click.

## Section 13

# Photographs and Pictures

# When to Use Photographs/Pictures

Use photographs when every visible aspect of the object is relevant.

Good uses:

- product appearance,
- people or places,
- physical damage evidence,
- real-world examples,
- visual identification.

Photographs communicate details that are hard to describe textually.

# Photograph Guidelines

- Prefer JPEG or modern photographic formats such as WebP/AVIF where supported.
- On the initial page, use a small version or thumbnail.
- Link to a larger, high-quality version if needed.
- Show the most relevant detail clearly.
- Use close-up shots with clean backgrounds.
- Avoid cluttered scenes with too many people or objects.
- Use people photographs carefully; test whether they increase trust or distract.

## Photographs/Pictures

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- Use:
    - When every aspect of the image is relevant.
  - Guidelines:
    - Use JPEG format.
    - On the initial page, display a small version.
      - Display a thumbnail size image.
      - Zoom in on most relevant detail.
      - Link to larger photos showing as much detail as needed.
    - Include fewer people and objects in less complicated settings than print.
    - Emphasize close-up shots with clean backgrounds.
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Figure 3: Photographs/Pictures guidelines: thumbnail and larger-photo strategy. Source: Galitz, Step 11, printed p. 676.

## Section 14

### Practical Design Application



## Example: Product Listing Page

| Design decision      | Good image practice                                |
|----------------------|----------------------------------------------------|
| Product image        | Use clear thumbnail with link to larger view       |
| Image size           | Compress and serve responsive sizes                |
| Label                | Provide product name below image                   |
| Alt text             | Describe product, not “image1.jpg”                 |
| Navigation           | Make clickable area obvious                        |
| Background           | Keep plain to preserve product visibility          |
| Internationalization | Avoid culturally specific symbols unless localized |

# Example: Learning Management System

For a course page, images may be used to:

- identify modules,
- show diagrams of concepts,
- provide screenshots of software steps,
- illustrate examples,
- support memory through visual association.

But images should not:

- replace essential instructions,
- make pages heavy,
- distract from learning tasks,
- lack descriptions for accessibility.

## Section 15

### Summary Tables

# Image Design Do's and Don'ts

| Do                              | Don't                                           |
|---------------------------------|-------------------------------------------------|
| Use images with clear purpose   | Add images only for decoration                  |
| Use familiar/standard images    | Invent unfamiliar visual language unnecessarily |
| Provide labels and alt text     | Assume everyone can see or understand the image |
| Keep images small and optimized | Upload oversized images directly                |
| Distinguish clickable images    | Make users guess what is interactive            |
| Use thumbnails for large images | Force large downloads immediately               |
| Test with users                 | Rely only on designer preference                |

# Complete Image Checklist

Before finalizing an interface image, verify:

- ☐ The image has a clear purpose.
- ☐ It conveys the intended message.
- ☐ It is familiar to target users.
- ☐ It is legible at actual display size.
- ☐ It is consistent with other interface images.
- ☐ It has descriptive text, label, or alt text.
- ☐ It is optimized for file size.
- ☐ The file format is appropriate.
- ☐ It does not hide important content above the fold.
- ☐ It does not reduce text readability.
- ☐ It works for accessibility needs.
- ☐ It is suitable for international users.

## Section 16

### Key Takeaways

# Final Message

Images are powerful only when they improve communication and task performance.

A good UI image is:

- meaningful,
- familiar,
- legible,
- consistent,
- accessible,
- efficiently loaded,
- culturally appropriate,
- supportive of the user's task.

**Conclusion:** In Software User Interface Design, image design is not about making screens look rich; it is about making interaction clearer, faster, and more usable.

## Section 17

### References



- Galitz, W. O. (2007). *The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques* (3rd ed.). Wiley.
- Galitz, Step 11: “Create Meaningful Graphics, Icons, and Images,” especially “Graphics,” “Images,” “Image Maps,” and “Photographs/Pictures,” pp. 669–677.
- Galitz, Step 10: “Web Page Accessibility Design,” especially image accessibility guidelines, pp. 642–644.